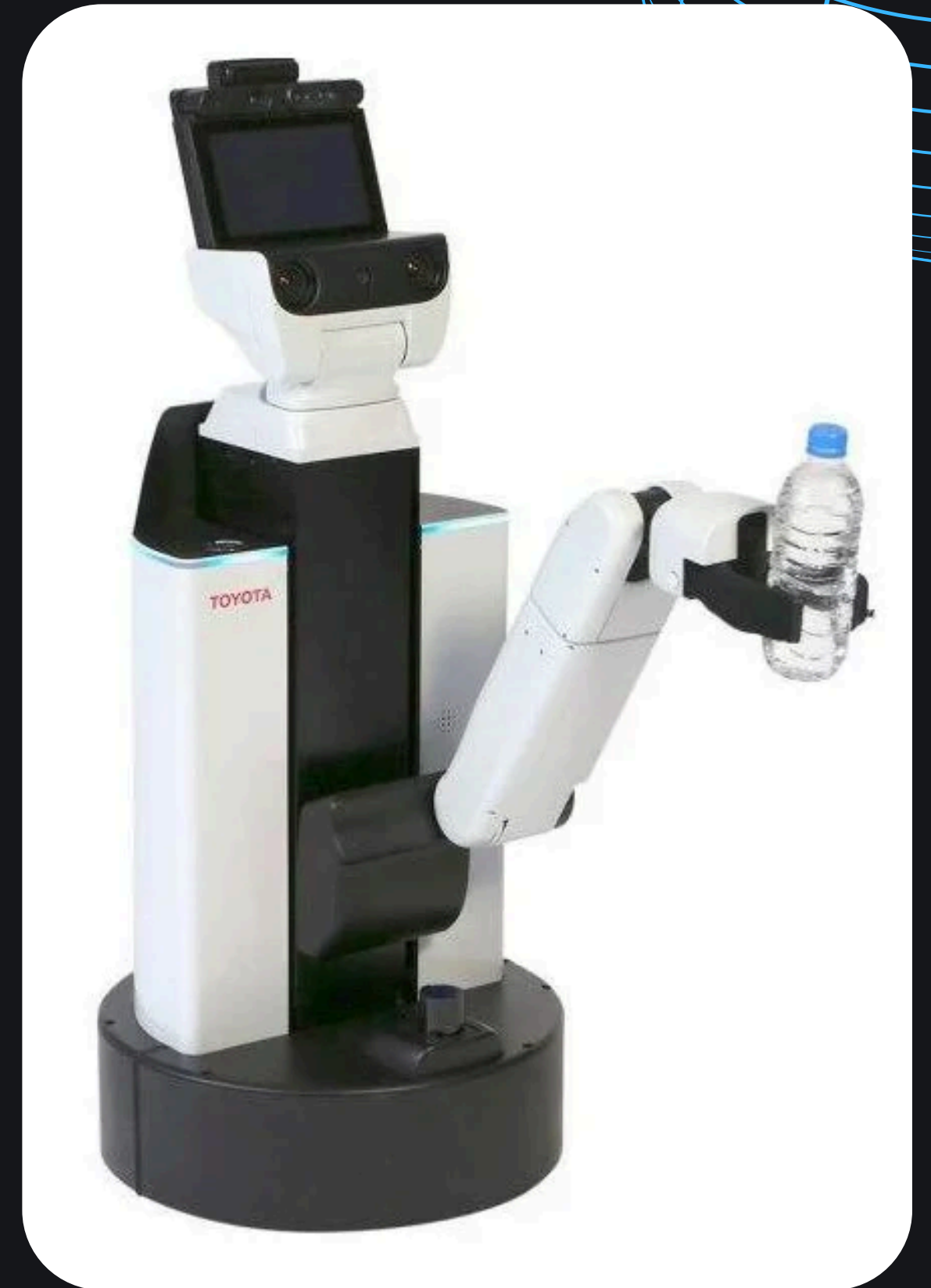


Instruction-Conditioned Robotic Grasping through Vision-Language Affordance Reasoning

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How Should a Robot Decide Where to Grasp an Object?

Humans naturally adapt their grasp to the intended task.
Most robotic grasp planners optimise for grasp stability
rather than task-specific intent.



1 Drink
"I'd like a hot drink"



Needs access to the rim.

2 Hand Over
"Pass it to me"



Handle must stay accessible.

3 Store / Move
"Put it away"



Task intention determines which surfaces can and cannot be grasped.

Research question

Can a robot reliably select the correct graspable part and physically lift the object, when perception is performed from realistic robot-mounted viewpoints?

System	Measures	Reported result
Contact-GraspNet (Sundermeyer et al., 2021)	Grasp success on unseen objects	>90%
AnyGrasp (Fang et al., 2023)	Objects cleared from a bin	93.30%
GLOVER (Ma et al., 2024)	Correct part named / grasp success	86.0% / 76.3%
ShapeGrasp (Li et al., 2024)	Correct part named	92%
AffordGrasp (Tang et al., 2025)	End-to-end task success	74%

Identified gap

No existing evaluation establishes whether a semantically selected part remains measurable and physically graspable from realistic robot-mounted viewpoints and whether that selection results in a verified physical lift.



Objectives & Contributions

01. Infer target object and part from intent alone, without a hand-written instruction-to-part table

13 scenes × 5 repeated passes, scored against intended object

02. Convert inferred task constraints into executable grasp parameters

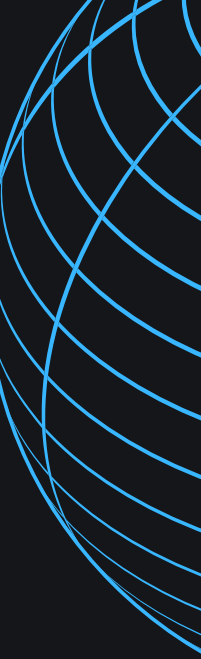
Deterministic mapping from constraint fields to gripper closure

03. Measure whether perception resolves the parts the reasoning names

Three grounding methods on the same real captures, plus range sweeps

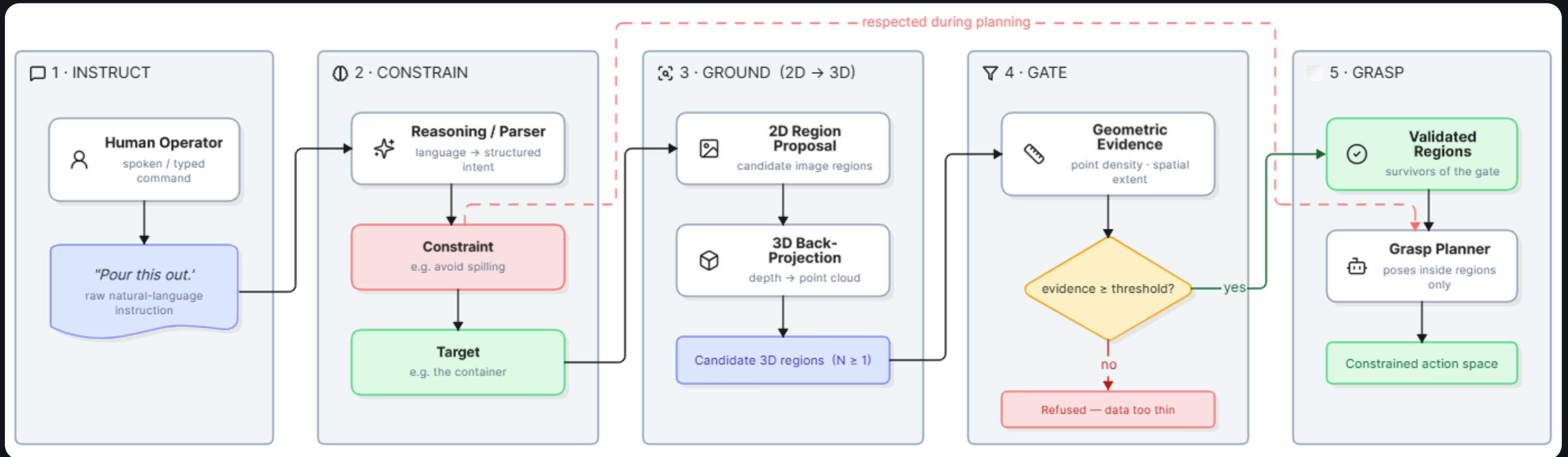
04. Execute the full pipeline on HSR hardware

11 logged trials, success defined as verified object rise



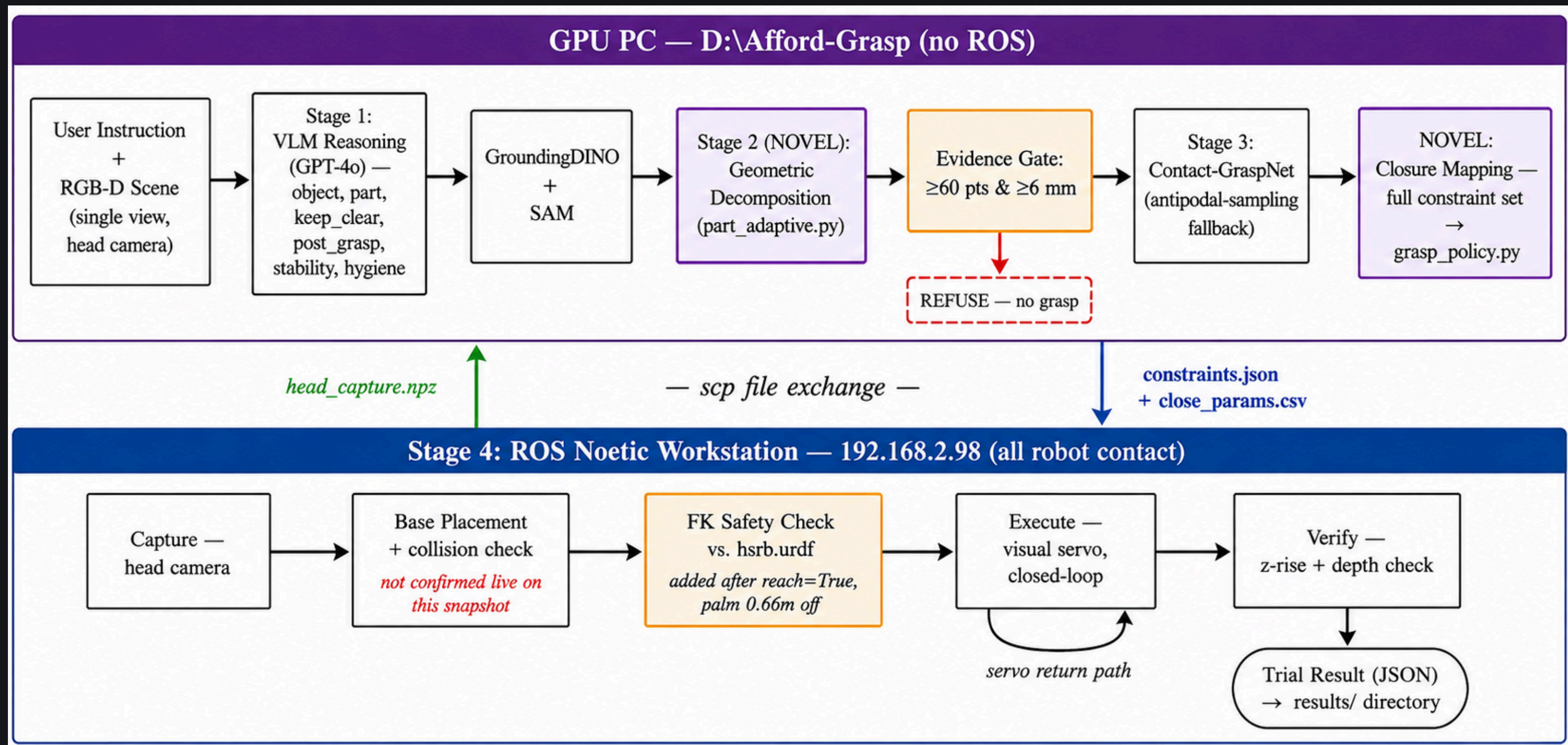
Conceptual Framework

Task intent progressively narrows the set of regions and actions that the robot is permitted to execute.

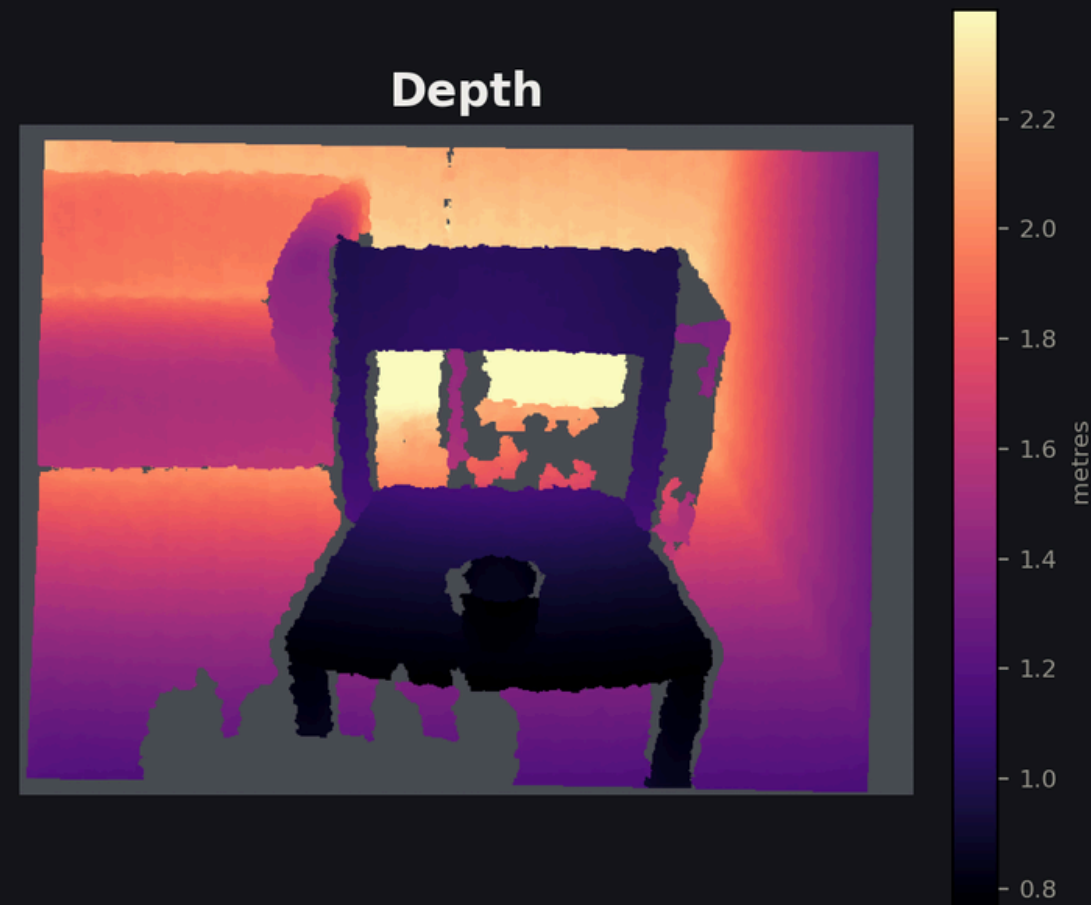


System Architecture

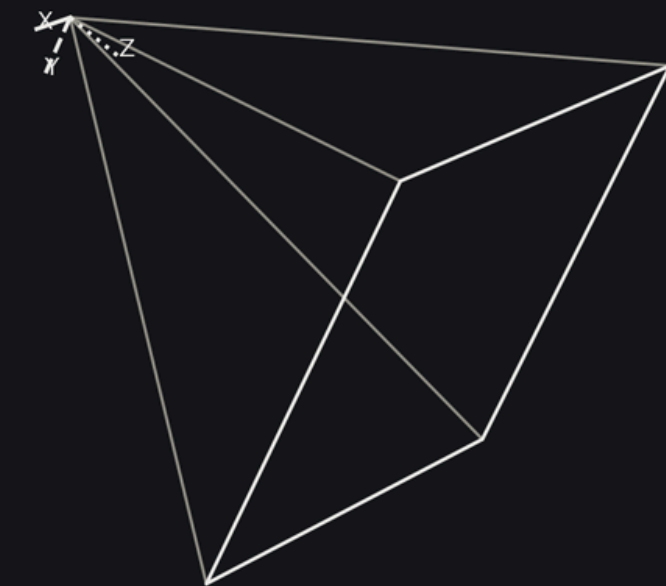
A two-machine architecture based on the conceptual framework, separates high-level perception and reasoning from robot-side feasibility, safety and execution.



Scene acquisition



Camera Frustum

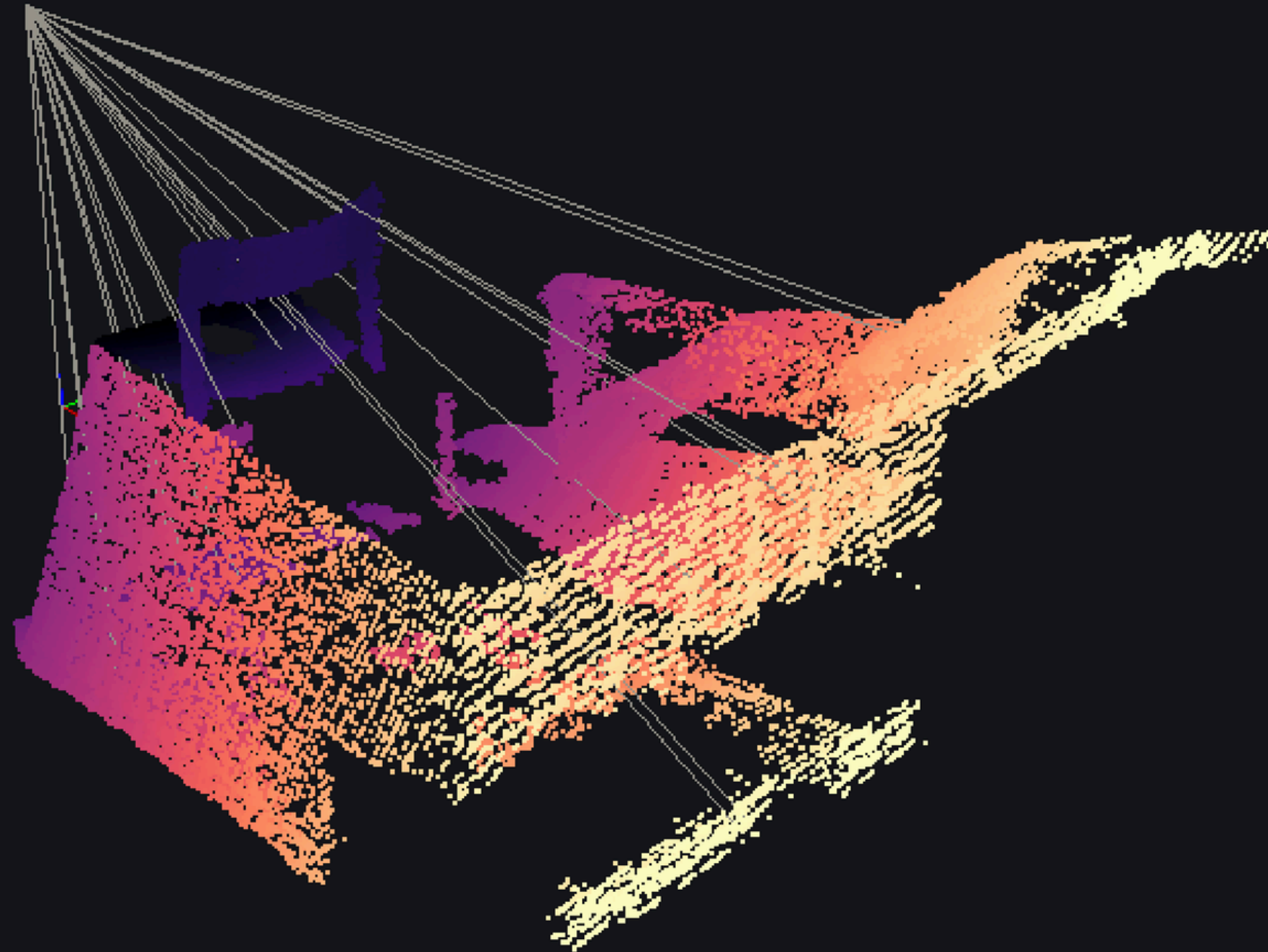


head_rgbd_sensor_rgb_frame
(schematic, head tilt -0.49 rad)

RGB-D observation of the object from the robot's perception system



3D geometric representation



Fused point cloud providing the geometry for downstream grasp reasoning



Task Instruction to Actionable Constraints

```
{  
  "instruction": "I'd like a hot drink",  
  "target_object": "red mug",  
  "target_part": "handle",  
  "constraints": {  
    "keep_clear": [  
      "rim"  
    ],  
    "keep_clear_reasons": [  
      "Covering this would block the liquid's  
      exit path during pouring."  
    ],  
    "post_grasp_motion": "tilt_pour",  
    "stability_priority": "high",  
    "thermal_or_hygiene": "hot surface"  
  }  
}
```



Grounded object mask:
3830 px

Bounding box:
(305,307)-(374,381)

Bounding box derived geometrically from the real object mask (grasps_out.npz); not a separately stored detection box.

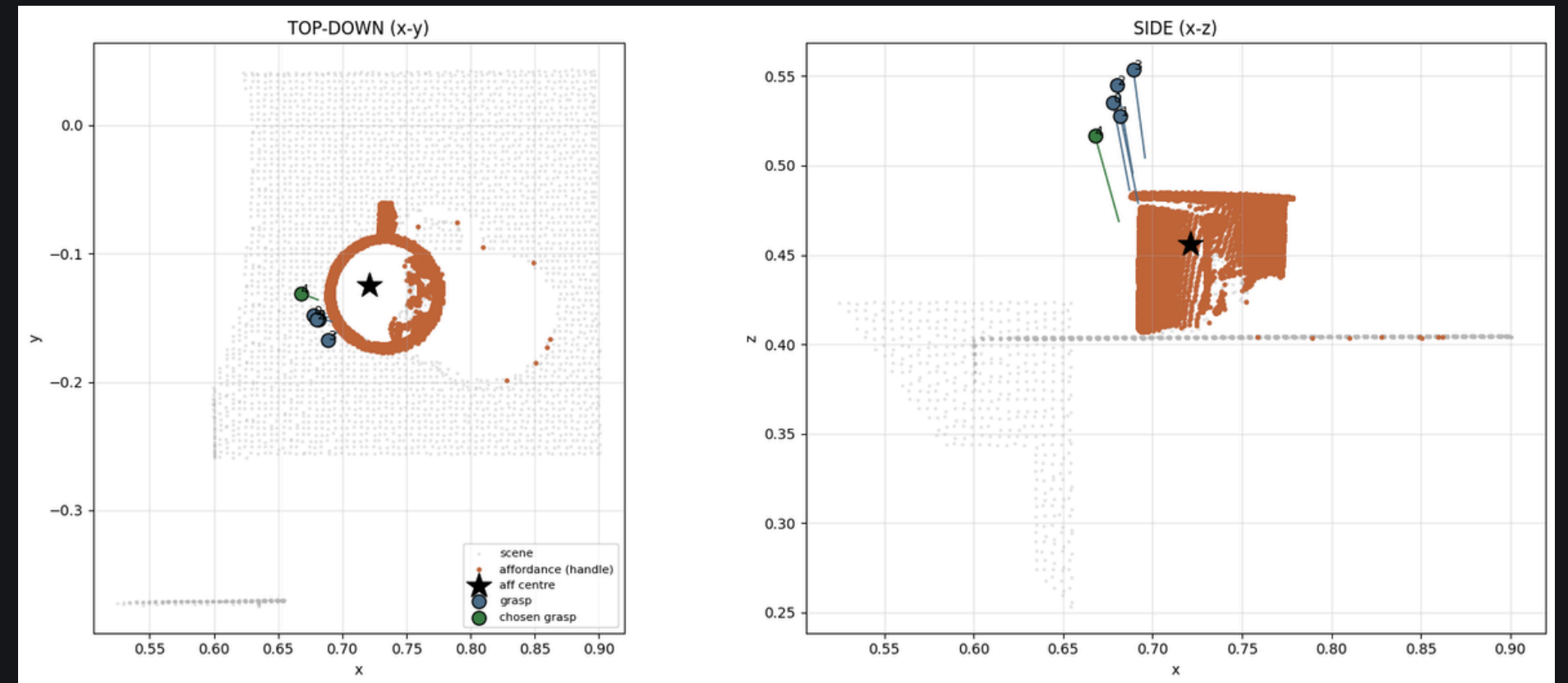
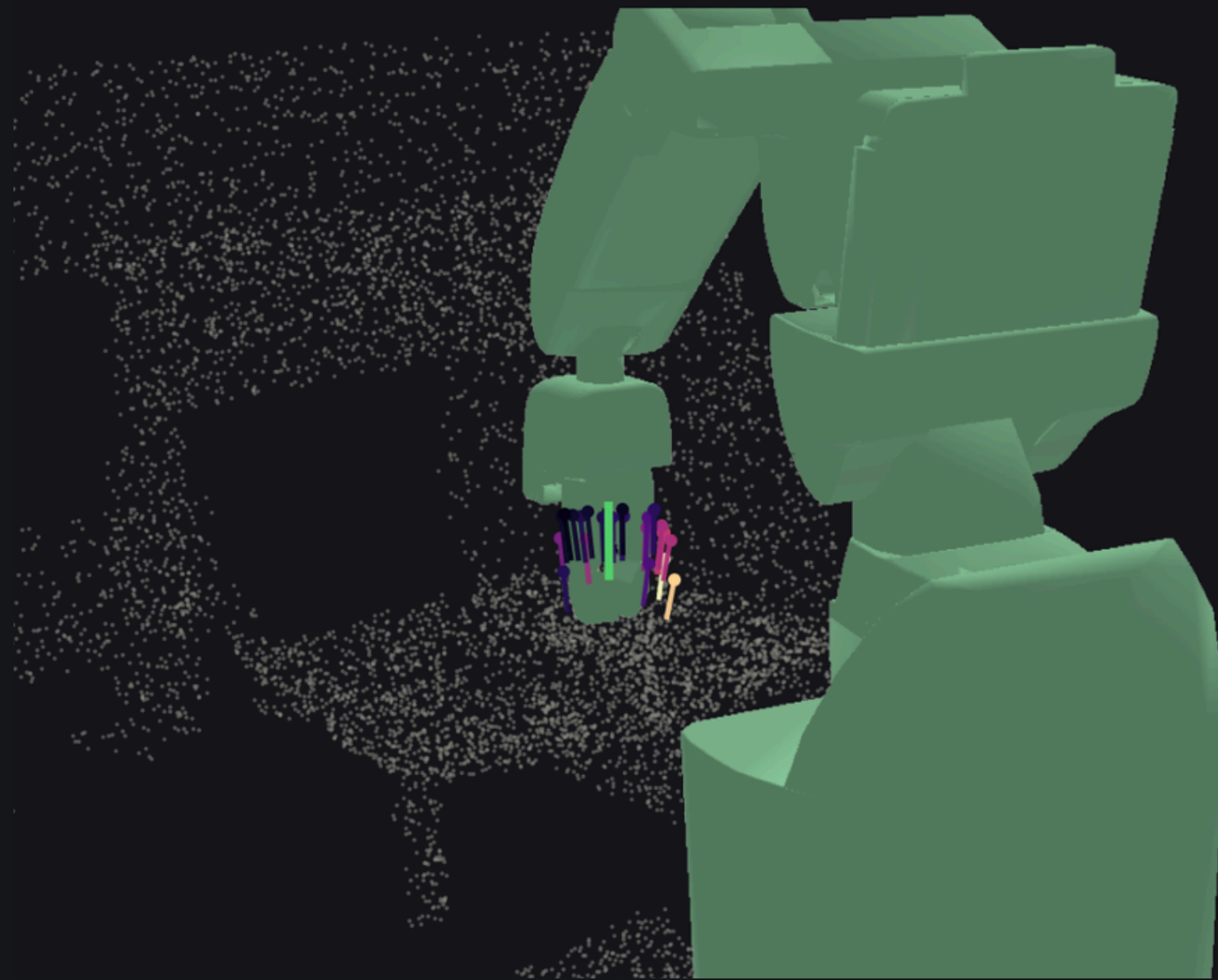


Part Decomposition

Observed geometry is decomposed into measurable structural components to ground the semantic target.



From Validated Affordance to Physical Grasp



25 real candidates (indices 0–24)
#24 (green) executed on the real robot

CGN generates candidate 6-DoF grasps from the observed object geometry, providing the initial grasp set for downstream feasibility filtering.

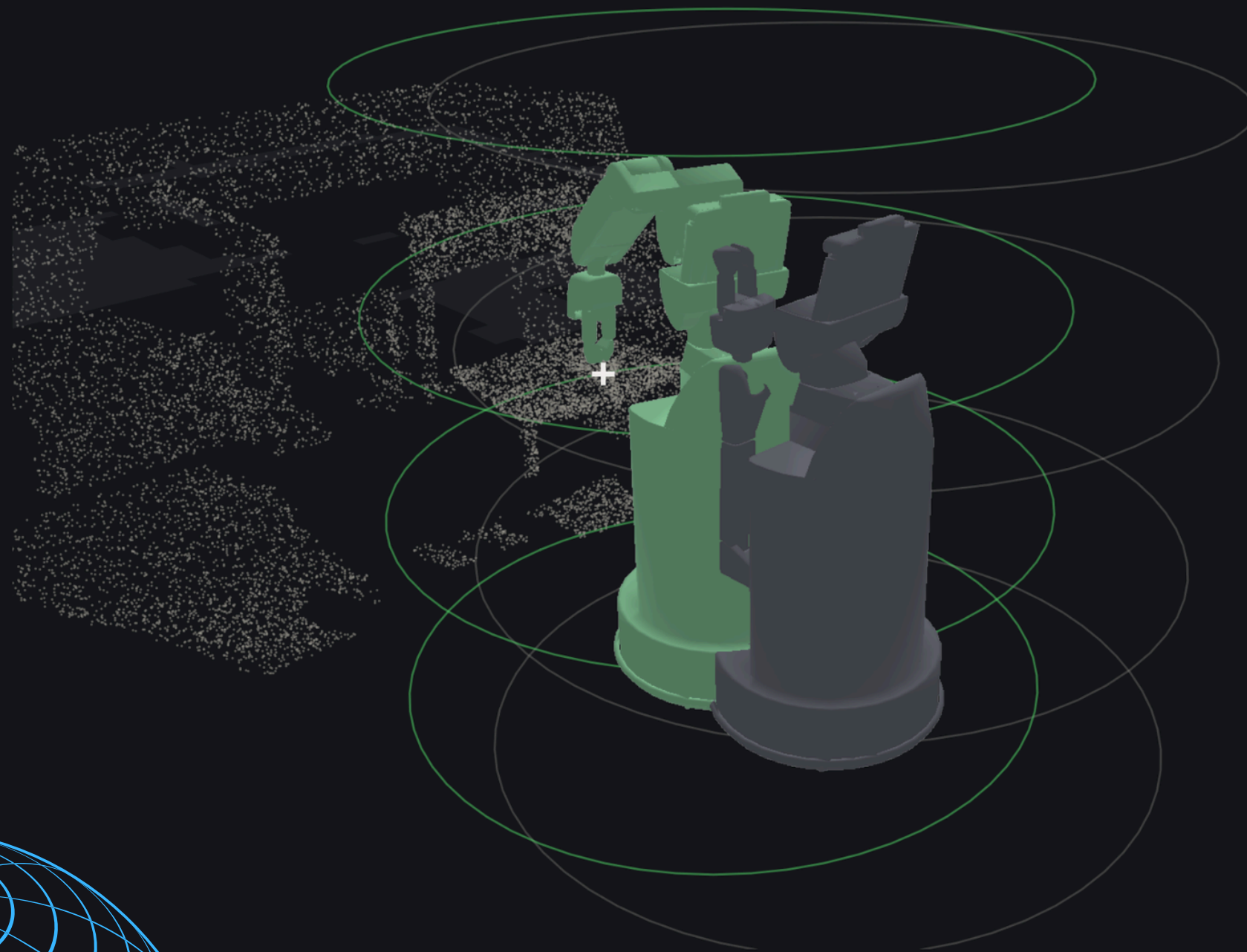


Feasibility-aware search identifies reachable, collision-free grasp configurations.



Grasp Generation & Execution

Collision-aware repositioning converts an unreachable target into a reachable grasp.



Grey robot + cylinder

initial base (stowed arm, not measured)

Green robot + cylinder

candidate #24 (real logged joints)

Dotted green line

real base repositioning

Faint squares

real keep-out cells (clear=0.32 m)

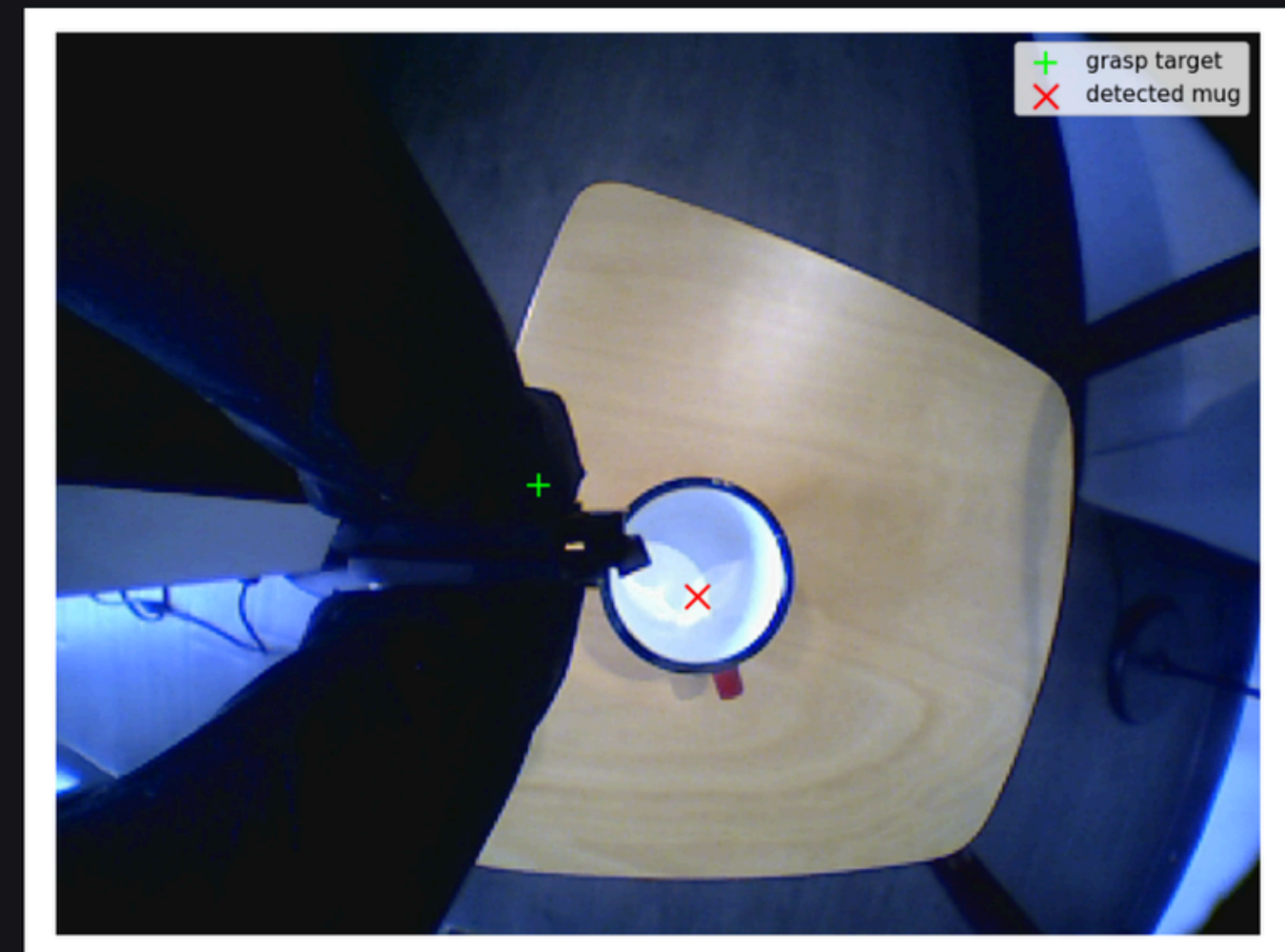
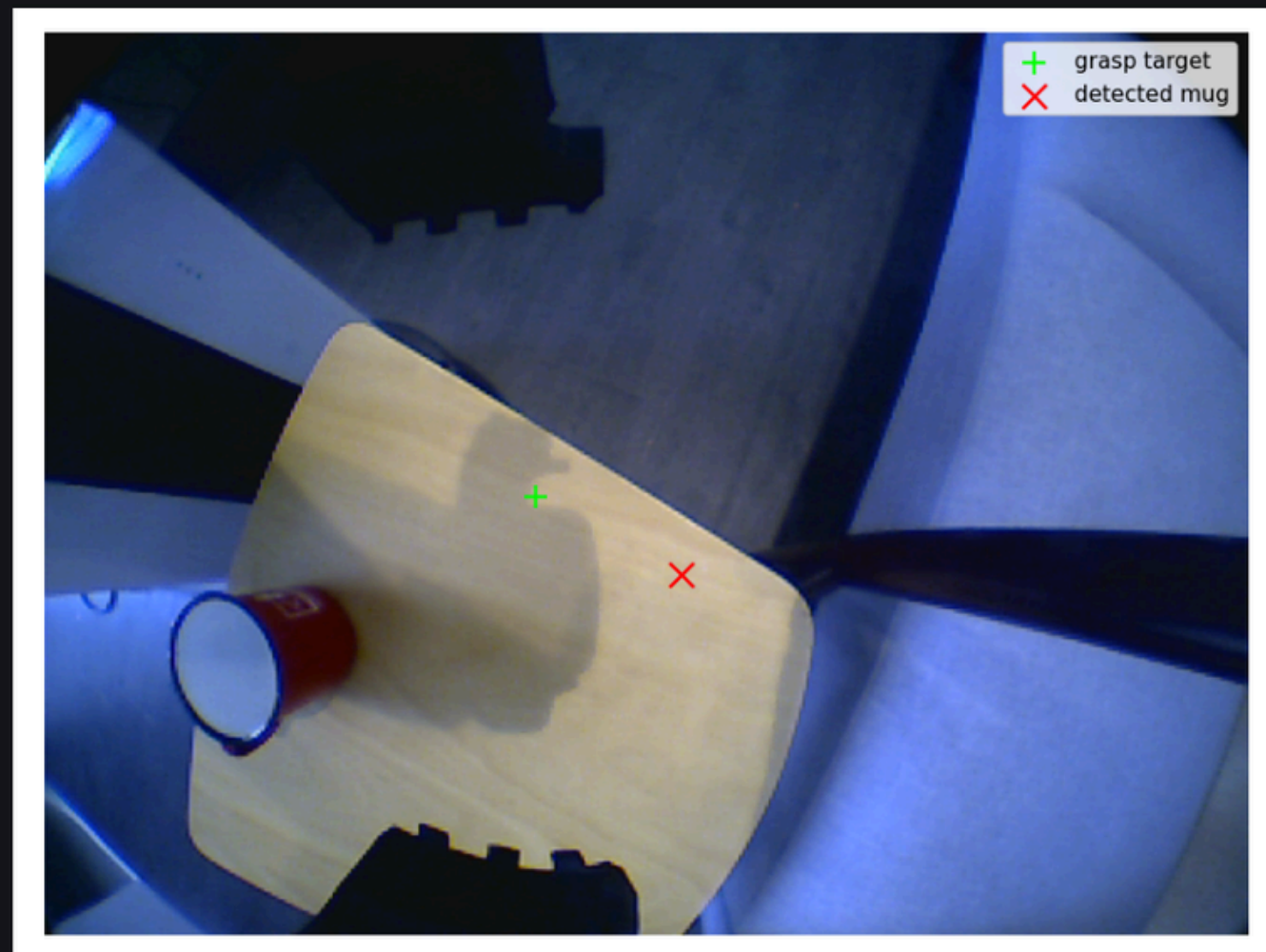
White +

real target (aff_center_3d)

Grasp Generation & Execution

Visual feedback closes the loop, reducing image-space error to achieve precise alignment.

MEASURED ERROR → EMPIRICAL JACOBIAN → CORRECTIVE MOTION → REDUCED ERROR



46.9px

before servo

-73%

12.7px

after servo

Grasp Generation & Execution

Execution is gated by pose consistency and successful lift verification, ensuring the generated grasp is valid before transport.

G2: PALM vs FK (grasp pose)



pos_err = 0.0001 m (< 0.02) ori_err = 0.02 deg (< 15.0) PASS

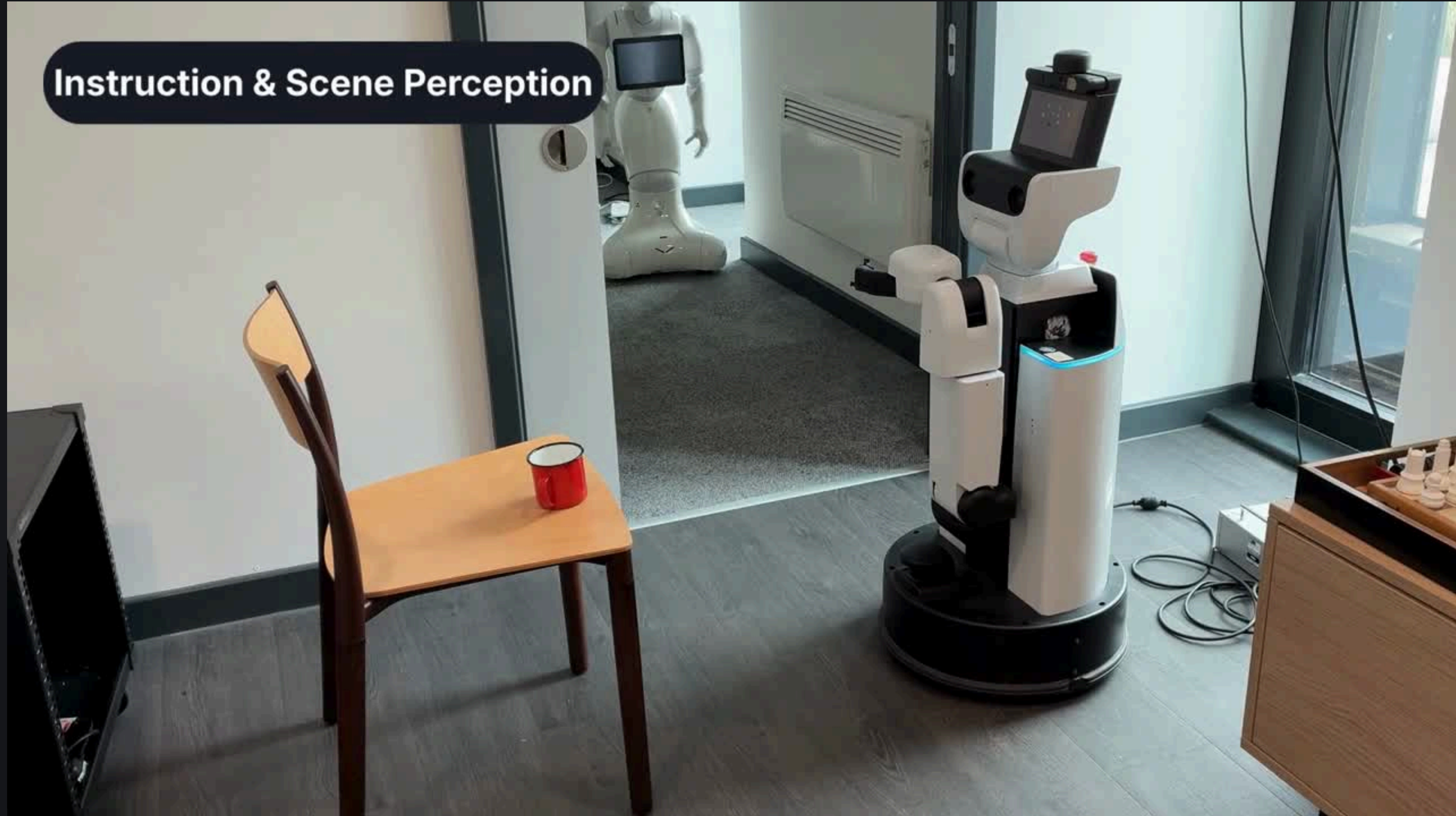
G3: LIFT-VERIFY (post-lift pose)



hand_motor = -0.3521 (> -0.885) z_rise = 0.1418 m lift_executed = yes PASS



Real-Robot Demonstration



The model gets the instruction and the scene,
and must name the object and part with no instruction-to-part table available to it.

61/65

correct

instruction: "I'd like a hot drink"
both scenes contain
a pot

cluster_mug_cokecan_pot

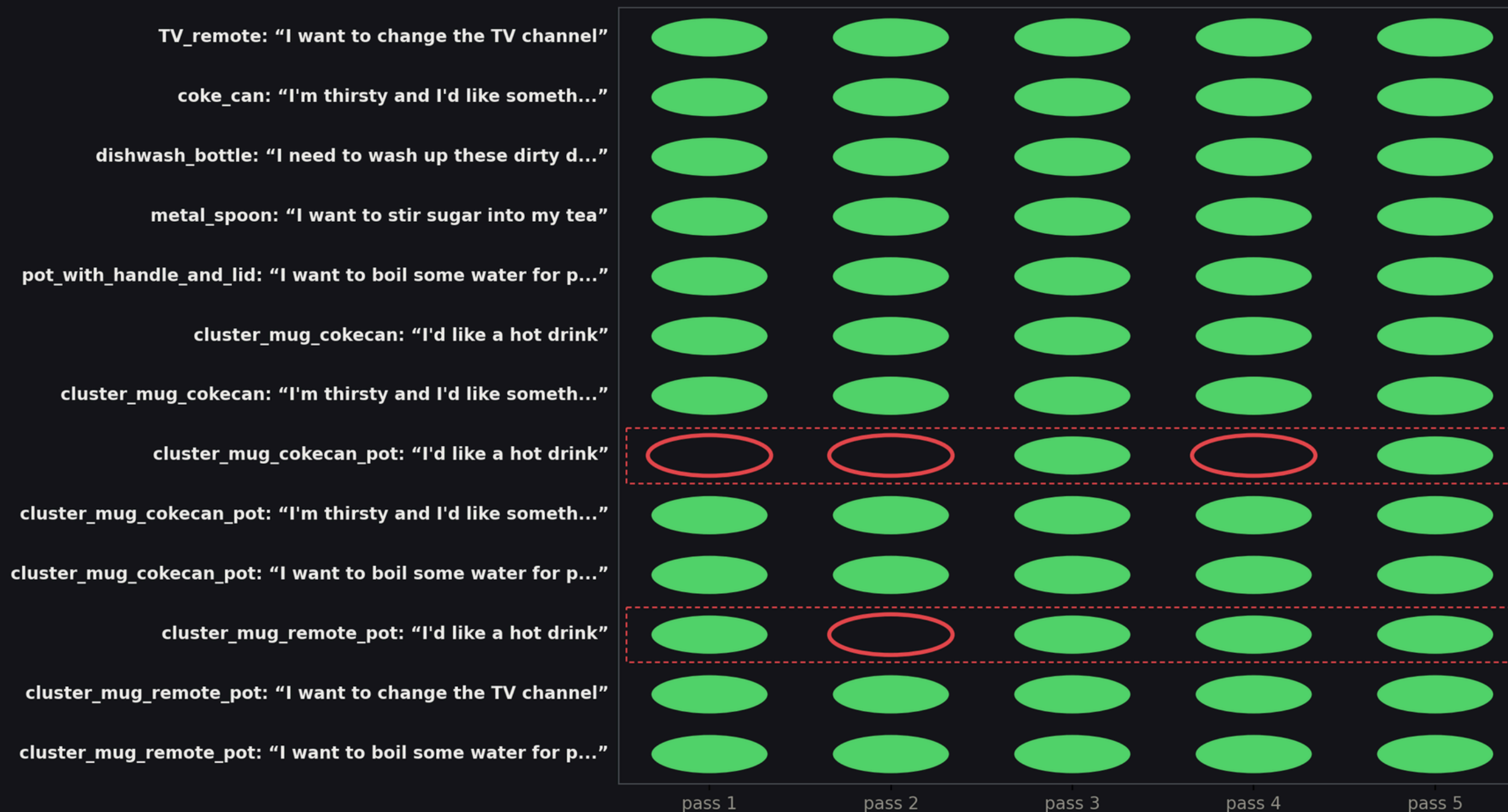


cluster_mug_remote_pot



Offline Reasoning Results

Intent identification was stable across repeated VLM passes, but clutter introduced specific failures.



Reasoning is generally stable, but the failures concentrate in ambiguous cluttered scenes.

Experimental Methodology

**The mask that comes back as the handle covers 97% of the whole-object detection box.
In other words, the system has not localised a part**



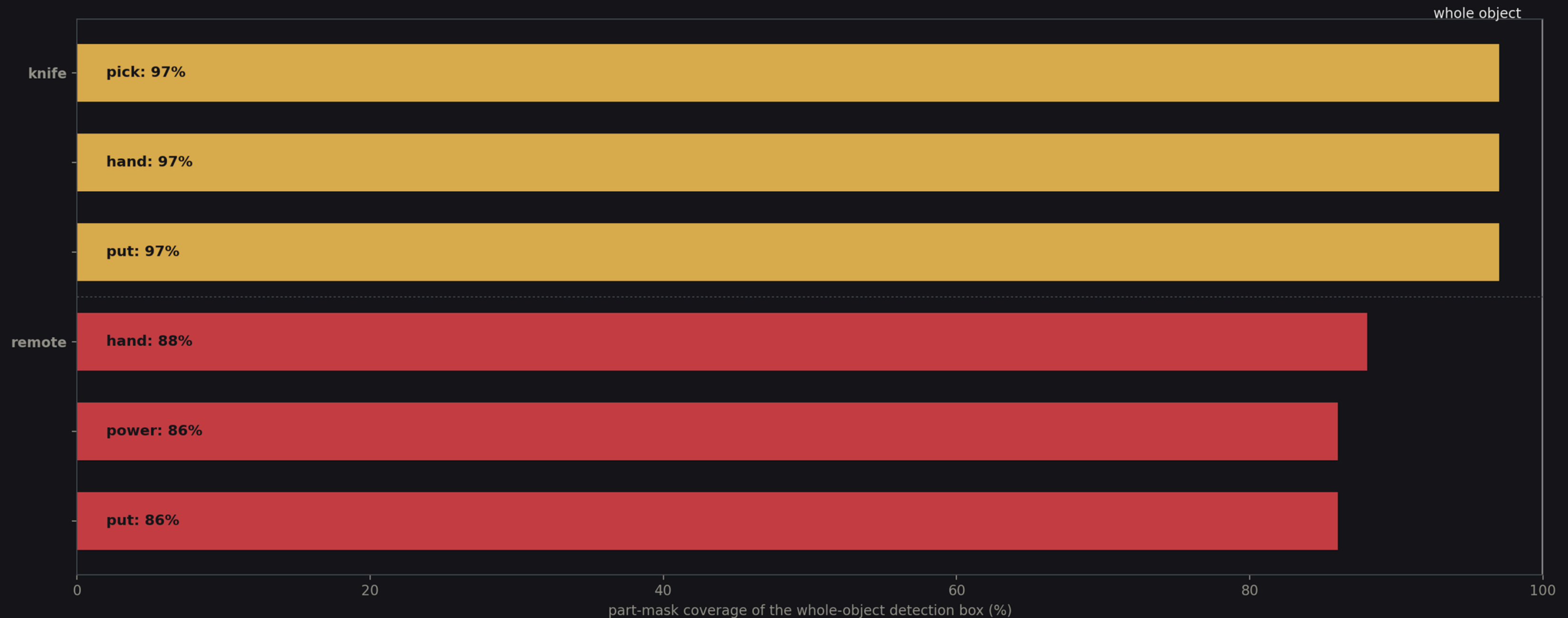
97%

**of the whole-object box
is what came back as
the "part" mask**



Perception Results

The VLM can name the intended part, but geometric grounding does not reliably recover that part.



Semantic identification is more reliable than geometric part grounding.

Critical Analysis & Lessons Learned

captures_pairs_2/cap_remote_head_near.npz -- dashed = pre-fix (wrong), solid = post-fix (correct)



query: "sides"

pre-fix: main_body -- 286pts/38mm -- evidence: OK (WRONG region)

post-fix: segment_b -- 1565pts/43mm -- evidence: OK (correct region)

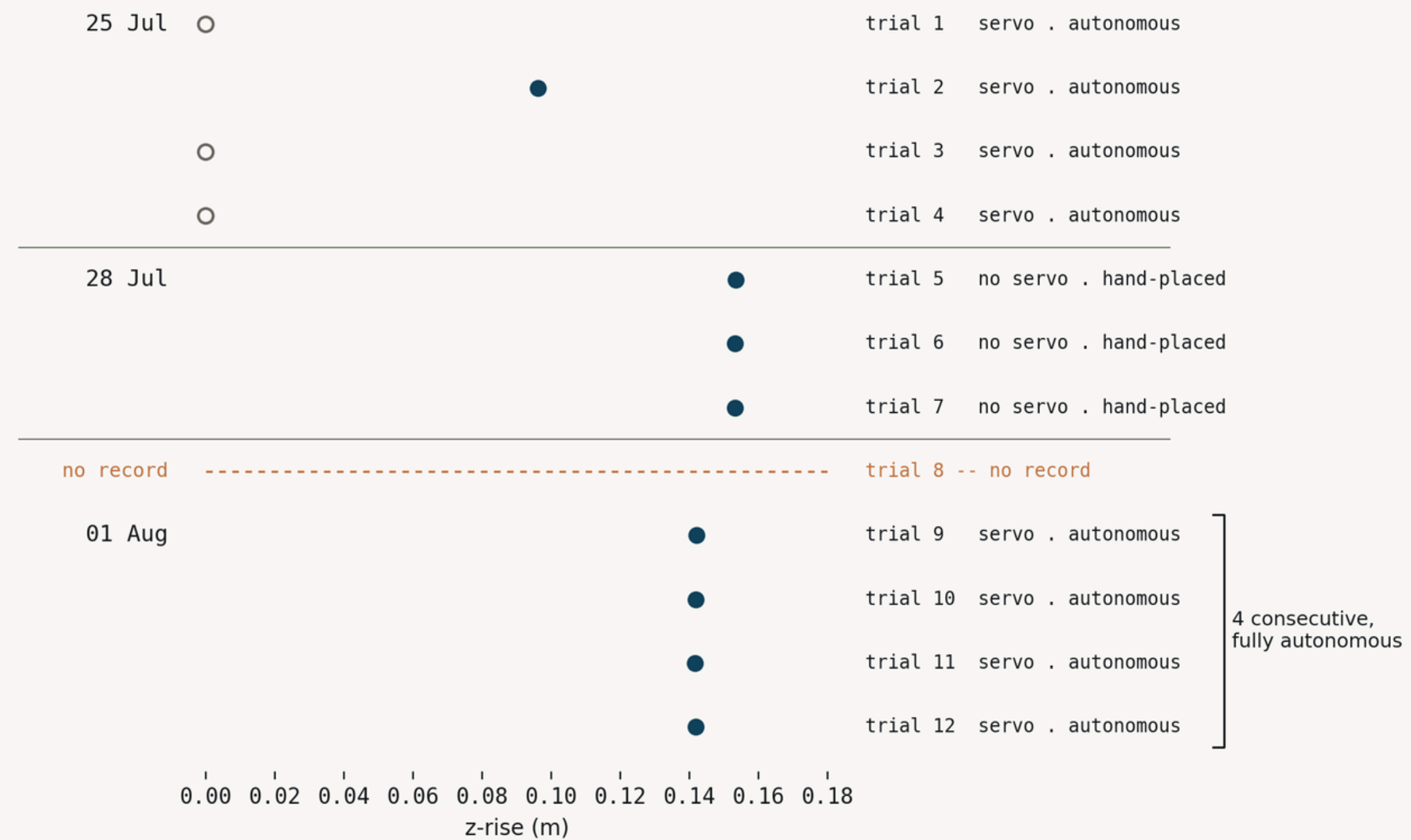
query: "middle section"

pre-fix: main_body -- 286pts/38mm -- evidence: OK (WRONG region)

post-fix: segment_b -- 1565pts/43mm -- evidence: OK (correct region)



Hardware Results



Experimental Methodology

CONFIRMED CHAIN A -- lift-fault (27-Jul patch, spot-verified)



CONFIRMED CHAIN B -- incident #10 (odometry guard, spot-verified)



UNCONFIRMED -- real logged failures, root cause not independently linked to a specific fix in this audit



Limitations

1 - Perception remains the dominant bottleneck

Part grounding is sensitive to segmentation quality and camera viewpoint.

Small or fragmented geometric components can fall below the evidence threshold even when the intended part is visually obvious.

The evidence gate verifies sufficient geometric evidence, but cannot verify that the selected component is semantically correct.

2 - VLM identification is not perfectly stable

Repeated intent-only identification achieved 61/65 correct predictions (94%) across 13 real scenes × 5 passes.

The remaining failures occurred in cluttered scenes containing a pot, indicating sensitivity to competing objects and scene context.

3 - Geometric decomposition is object-dependent

The current decomposition relies on shape-adaptive heuristics such as protrusion detection and clustering.

Complex objects, fragmented parts, occlusions and unusual geometries can produce ambiguous or incomplete components.

The current system therefore cannot guarantee a valid part decomposition for every object.

4 - Execution is constrained by the physical platform

Real-world execution remains sensitive to calibration, odometry, grasp correction and actuator behaviour.

Documented execution failures required recovery and targeted fixes.

Successful grasp generation does not guarantee successful physical execution.



Future Work

1 — Stronger part grounding

- Replace object-level re-localisation with dedicated part-aware segmentation.
- Fuse RGB, depth and 3D geometry to obtain more reliable part boundaries.
- Introduce confidence estimates that distinguish semantic correctness from geometric sufficiency.

2 — More robust part reasoning

- Learn object-part relationships rather than relying primarily on fixed relation hints and geometric heuristics.
- Use multiple grounding hypotheses and cross-check them before grasp generation.
- Explicitly represent uncertainty when the intended part cannot be reliably identified.

3 — Larger and more systematic evaluation

- Expand beyond the current real-object and clutter scenes.
- Evaluate more object categories, viewpoints, lighting conditions and clutter configurations.
- Measure perception, grasp quality and execution success separately to identify where improvements actually occur.

4 — Closed-loop task-aware manipulation

- Extend the current Reason → Ground → Grasp → Execute pipeline towards continuous visual feedback.
- Re-estimate the target after approach and after contact rather than treating perception as a one-shot decision.
- Allow the robot to recover by re-grounding or selecting an alternative grasp when execution conditions change.



Conclusion

The system selects correctly and lifts reliably but the part it names is not the part it can measure.

61/65	86–97%	08/08/2026
intent-only passes named the correct object and part (94%, 13 scenes × 5)	of the whole-object box is what came back as the "part" mask	logged trials reached a verified lift 4 consecutive fully autonomous
REASONING — reliable	PERCEPTION — does not resolve the named part	EXECUTION — achievable

**The reasoning layer expresses finer distinctions than the perception layer can resolve.
A system built this way will report part-level success while grasping at the object level
and no existing evaluation would detect it.**

